

Math 242, S15
Exam 2

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	18	
2	20	
3	24	
4	18	
5	20	
Total	100	

Good Luck !

Problem 1. (18 points) Given the following homogeneous second-order differential equation:

$$y'' + 4y' + 13y = 0.$$

(a) Find the general solution of the given differential equation.

Solution: Characteristic equation

$$r^2 + 4r + 13 = 0$$

$$\Rightarrow r = \frac{-4 \pm \sqrt{4^2 - 4 \cdot 13}}{2} = -2 \pm 3i$$

$$\Rightarrow y(x) = e^{-2x}(C_1 \cos 3x + C_2 \sin 3x)$$

(b) Find a particular solution of the equation that satisfies the initial conditions

$$y(0) = 3, \quad y'(0) = 0.$$

$$\begin{aligned} \text{Solution: } y'(x) &= e^{-2x}(-3C_1 \sin 3x + 3C_2 \cos 3x) \\ &\quad + (-2)e^{-2x}(C_1 \cos 3x + C_2 \sin 3x) \\ &= e^{-2x}[(3C_2 - 2C_1)\cos 3x + (-3C_1 - 2C_2)\sin 3x] \end{aligned}$$

$$y(0) = 3 \Rightarrow C_1 = 3$$

$$y'(0) = 0 \Rightarrow 3C_2 - 2C_1 = 0 \Rightarrow 3C_2 = 6 \Rightarrow C_2 = 2$$

Thus

$$y(x) = e^{-2x}(3\cos 3x + 2\sin 3x)$$

Problem 2. (20 points) Given a nonhomogeneous linear third-order differential equation:

$$y^{(3)} - 6y'' + 9y' = 6x + 5.$$

(a) Find the complementary solution y_c of the above nonhomogeneous equation (i.e., the general solution of the associated homogeneous equation).

Solution: Characteristic equation

$$r^3 - 6r^2 + 9r = 0$$

$$r(r-3)^2 = 0$$

$$r_1 = 0, r_2 = r_3 = 3$$

$$\Rightarrow y_c(x) = C_1 + e^{3x}(C_2 + C_3x)$$

(b) Then find a particular solution y_p of the nonhomogeneous equation.

Solution $y_p(x) = x^s(AX+B)$ $s=1$
 $= AX^2 + BX$ ←

$$y_p'(x) = 2Ax + B, y_p''(x) = 2A, y_p^{(3)}(x) = 0$$

$$\begin{aligned}\Rightarrow y_p^{(3)} - 6y_p'' + 9y_p' &= -6(2A) + 9(2Ax + B) \\ &= -12A + 18Ax + 9B \\ &= 18Ax + (-12A + 9B) \equiv 6x + 5\end{aligned}$$

$$\Rightarrow \begin{cases} 18A = 6 \\ -12A + 9B = 5 \end{cases} \Rightarrow \begin{cases} A = \frac{1}{3} \\ B = 1 \end{cases}$$

$$\text{Thus } y_p(x) = \frac{1}{3}x^2 + x$$

Problem 3. (24 points) Find Laplace or inverse Laplace transforms of the functions below:

(a) $\mathcal{L}\{t^4 - 2\cos(3t)\}$

$$= \frac{4!}{s^5} - \frac{2s}{s^2+9}$$

$$= \frac{24}{s^5} - \frac{2s}{s^2+9}$$

(b) $\mathcal{L}\{t \cosh(4t)\}$ using the formula $\mathcal{L}\{-tf(t)\} = F'(s)$ where $F(s) = \mathcal{L}\{f(t)\}$.

$$\mathcal{L}\{t \cosh(4t)\} = \frac{d}{ds} \left[\frac{s}{s^2-16} \right]$$

$$= - \frac{1 \cdot (s^2-16) - s \cdot (2s)}{(s^2-16)^2}$$

$$= \frac{s^2+16}{(s^2-16)^2}$$

(c) $\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+16)}\right\}$ using the formula $\mathcal{L}^{-1}\{F(s)G(s)\} = f(t) * g(t)$ where $\mathcal{L}^{-1}\{F(s)\} = f(t)$ and $\mathcal{L}^{-1}\{G(s)\} = g(t)$.

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+16)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} * \mathcal{L}^{-1}\left\{\frac{1}{s^2+16}\right\}$$

$$= 1 * \frac{1}{4} \sin 4t$$

$$= \int_0^t \frac{1}{4} \sin 4(t-\tau) d\tau$$

$$= \frac{1}{16} \cos 4(t-\tau) \Big|_{\tau=0}^{\tau=t}$$

$$= \frac{1}{16} (1 - \cos 4t)$$

Problem 4. (18 points) Use Laplace transform to solve the initial value problem:

$$x'' - 16x = 0; \quad x(0) = 3, \quad x'(0) = 6$$

Solution: Let $\mathcal{L}\{x(t)\} = X(s)$, then

$$\begin{aligned}\mathcal{L}\{x''(t)\} &= s^2 X(s) - x(0)s - x'(0) \\ &= s^2 X(s) - 3s - 6\end{aligned}$$

Then

$$\mathcal{L}\{x'' - 16x\} = 0$$

$$[s^2 X(s) - 3s - 6] - 16X(s) = 0$$

$$(s^2 - 16)X(s) = 3s + 6$$

$$X(s) = \frac{3s + 6}{s^2 - 16}$$

$$\begin{aligned}\Rightarrow x(t) &= \mathcal{L}^{-1}\left\{\frac{3s + 6}{s^2 - 16}\right\} \\ &= \mathcal{L}^{-1}\left\{\frac{3s}{s^2 - 16}\right\} + \mathcal{L}^{-1}\left\{\frac{6}{s^2 - 16}\right\} \\ &= 3 \cosh(4t) + \frac{3}{2} \sinh(4t)\end{aligned}$$

Problem 5. (20 points) Use Laplace transform to solve the initial value problem

$$x'' + 4x' - 5x = 2; \quad x(0) = x'(0) = 0.$$

Solution. Let $\mathcal{L}\{x(t)\} = X(s)$. then

$$\mathcal{L}\{x'(t)\} = sX(s) - x(0) = sX(s)$$

$$\mathcal{L}\{x''(t)\} = s^2X(s) - sx(0) - x'(0) = s^2X(s)$$

Then

$$\mathcal{L}\{x'' + 4x' - 5x\} = \mathcal{L}\{2\}$$

$$s^2X(s) + 4sX(s) - 5X(s) = \frac{2}{s}$$

$$(s^2 + 4s - 5)X(s) = \frac{2}{s} \Rightarrow X(s) = \frac{2}{s(s^2 + 4s - 5)}$$

$$\begin{aligned} \frac{2}{s(s^2 + 4s - 5)} &= \frac{2}{s(s-1)(s+5)} = \frac{A}{s} + \frac{B}{s-1} + \frac{C}{s+5} \\ &= \frac{A(s-1)(s+5) + Bs(s+5) + Cs(s-1)}{s(s-1)(s+5)} \end{aligned}$$

$$\text{let } s=0 \Rightarrow -5A = 2 \Rightarrow A = -\frac{2}{5}$$

$$s=1 \Rightarrow B \cdot (1) \cdot (6) = 2 \Rightarrow B = \frac{1}{3}$$

$$s=-5 \Rightarrow C(-5)(-6) = 2 \Rightarrow C = \frac{1}{15}$$

Thus

$$\begin{aligned} x(t) = \mathcal{L}^{-1}\left\{\frac{2}{s(s^2 + 4s - 5)}\right\} &= \mathcal{L}^{-1}\left\{-\frac{2}{5} \frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{3} \frac{1}{s-1}\right\} \\ &\quad + \mathcal{L}^{-1}\left\{\frac{1}{15} \frac{1}{s+5}\right\} \end{aligned}$$

$$= -\frac{2}{5} + \frac{1}{3}e^t + \frac{1}{15}e^{-5t}$$